

AWS Systems Manager Based Disk Utilization Discovery for EC2 Instances

Submitted By: Kiran Manjunath Raykar

1. Introduction

Organizations running workloads on Amazon EC2 often manage multiple instances across different environments. Monitoring disk utilization is essential to ensure application availability, prevent storage-related issues, and support capacity planning. Traditionally, administrators would connect to each server using SSH or RDP to check storage utilization, which becomes difficult and time-consuming at scale.

AWS Systems Manager provides a secure and centralized method to manage EC2 instances without requiring direct server access. Using Systems Manager Run Command, administrators can remotely execute commands and retrieve operational information from EC2 instances.

This document demonstrates how disk utilization can be retrieved from an EC2 instance using AWS Systems Manager, explains the AWS components involved, highlights benefits and limitations, and provides implementation screenshots and customer communication examples.

2. Objective

The objective of this exercise is to:

- Retrieve disk utilization information from an EC2 instance using AWS Systems Manager.
- Avoid direct SSH/RDP access to servers.
- Understand the AWS resources involved in the process.
- Demonstrate secure remote command execution.
- Evaluate benefits and limitations of the solution.
- Explore alternate approaches for collecting disk utilization information.

3. Solution Architecture

[Insert Architecture Diagram]

Administrator

|

v

AWS Systems Manager

|

v

Run Command

|

v

SSM Agent

|

v

EC2 Instance

|

v

df -h Command

|

v

Disk Utilization Output

4. AWS Components Used

4.1 Amazon EC2

Amazon EC2 provides the virtual server on which the workload is running. Disk utilization is collected from this instance.

4.2 AWS Systems Manager

AWS Systems Manager is a management service that enables administrators to remotely execute commands and manage EC2 instances without requiring SSH or RDP access.

4.3 SSM Agent

The SSM Agent runs inside the EC2 instance and communicates with AWS Systems Manager. It receives commands, executes them locally, and returns the results.

4.4 IAM Role

An IAM role was attached to the EC2 instance to grant permissions required for Systems Manager communication.

Role Used:

EC2-SSM-Role

Policy Attached:

AmazonSSMManagedInstanceCore

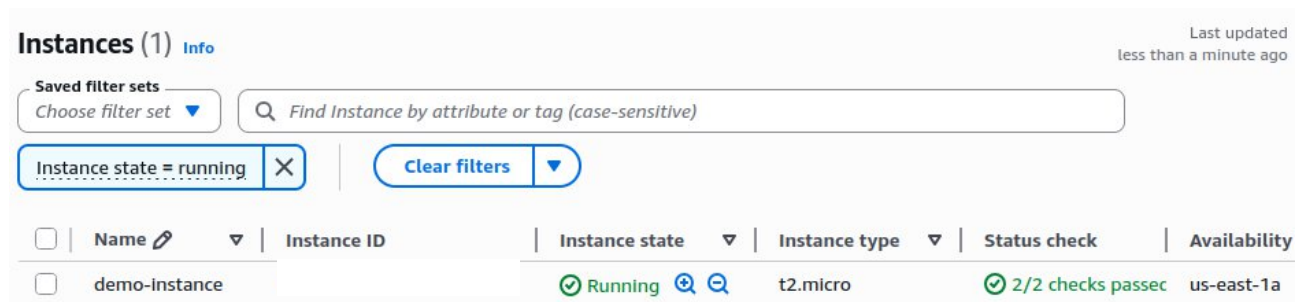
5. Implementation Steps

Step 1: Launch EC2 Instance

Created an Amazon Linux EC2 instance for testing.

Screenshot 1

EC2 Instance Running



The screenshot displays the AWS Management Console's 'Instances' page. At the top, it shows 'Instances (1)' with an 'Info' link. Below this, there are filter controls including 'Saved filter sets' and a search bar. A filter is applied: 'Instance state = running'. The main table lists one instance, 'demo-instance', which is in the 'Running' state, using the 't2.micro' instance type. The status check shows '2/2 checks passed', and the availability zone is 'us-east-1a'.

Instances (1) Info							Last updated less than a minute ago
Saved filter sets Choose filter set ▼		Find Instance by attribute or tag (case-sensitive)					
Instance state = running X		Clear filters ▼					
☐	Name ✎ ▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability	
☐	demo-instance		Running 🔍 🔍	t2.micro	2/2 checks passed	us-east-1a	

Step 2: Create IAM Role

Created IAM role named EC2-SSM-Role.

Attached policy:

AmazonSSMManagedInstanceCore

Screenshot 2

IAM Role Created [EC2-SSM-Role]

Roles (4) Info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you

☐	Role name	Trusted entities	Last activity
☐	AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (Se	6 minutes ago
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Linker	-
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service	-
☐	EC2-SSM-Role	AWS Service: ec2	10 minutes ago

Step 3: Attach IAM Role to EC2 Instance

Attached the IAM role to the EC2 instance.

Screenshot 3:

IAM Role Attached to EC2

Instance summary for i-04628...9 (demo-instance) Updated 1 minute ago

Instance ID
 i-04628

IPv6 address
-

Hostname type
IP name: ip-internal

Answer private resource DNS name
-

Auto-assigned IP address
 [Public IP]

IAM role
 [EC2-SSM-Role](#)

Public IPv
 54.19

Instance s
 **Runnin**

Private IP
 ip-10

Instance t
[t2.micro](#)

VPC ID
 vpc-0

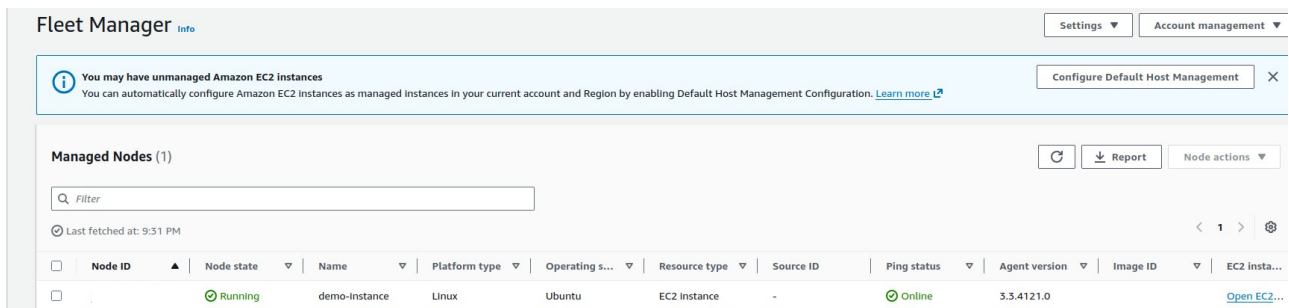
Subnet ID
 subne

Step 4: Verify Systems Manager Registration

Verified that the EC2 instance was successfully registered in AWS Systems Manager Fleet Manager.

Screenshot 4

Systems Manager / Managed Node



The screenshot shows the AWS Systems Manager Fleet Manager console. At the top, there's a header with 'Fleet Manager' and 'Info'. Below the header, there's a notification banner stating 'You may have unmanaged Amazon EC2 instances' with a 'Configure Default Host Management' button. The main section is titled 'Managed Nodes (1)' and contains a table with one node. The table has columns for Node ID, Node state, Name, Platform type, Operating s..., Resource type, Source ID, Ping status, Agent version, Image ID, and EC2 insta... The node listed is 'demo-Instance' with a 'Running' state and 'Online' ping status.

Node ID	Node state	Name	Platform type	Operating s...	Resource type	Source ID	Ping status	Agent version	Image ID	EC2 insta...
	Running	demo-Instance	Linux	Ubuntu	EC2 Instance	-	Online	3.3.4121.0		Open EC2...

Step 5: Execute Disk Utilization Command

Opened Systems Manager Run Command.

Selected document:

AWS-RunShellScript

Executed command:

df -h

Screenshot 5

Run Command Success

[AWS Systems Manager Run Command executed successfully on the EC2 instance]

Command ID: c8f84d96-fc30-417e-80c7-90f137567028

Cancel command

Retu

Command status

Overall status

Success

Detailed status

Success

targets

1

completed

1

error

0

delivery ti

0

Targets and outputs

Q Search command Invocations

Instance ID

Instance name

Status

Detailed Status

Start time

Finish time

-165.ec2.Internal

Success

Success

Thu, 11 Jun 2026 15:42:19 GMT

Thu, 11 Jun 2026 15:4

► CloudWatch alarm

Step 6: Review Output

Collected disk utilization output returned by Systems Manager.

Screenshot 6

df -h Output

Output on i-046281

Step 1 - Command description and status

Status

Success

Detailed status

Success

Response code

0

Step name

aws:runShellScript

Start time

Thu, 11 Jun 2026 15:42:19 GMT

Finish time

Thu, 11 Jun 2026 15:42:19 GMT

▼ Output

The command output displays a maximum of 24,000 characters. You can view the complete command output in either Amazon S3 or CloudWatch Logs, if you specify an S3 bucket or a logs group when you run the command.

Copy

Download

Filesystem Size Used Avail Use% Mounted on

/dev/root 6.8G 3.2G 3.6G 48% /

tmpfs 479M 0 479M 0% /dev/shm

tmpfs 192M 880K 191M 1% /run

tmpfs 5.0M 0 5.0M 0% /run/lock

► Error

=====

6. Results and Analysis

Command Executed:

df -h

Output Summary:

Filesystem Size Used Avail Use% Mounted on
/dev/root 6.8G 3.2G 3.6G 48% /

```
/dev/xvda16 881M 152M 667M 19% /boot  
/dev/xvda15 105M 6.1M 99M 6% /boot/efi
```

Analysis:

- Root Volume Size: 6.8 GB
- Used Space: 3.2 GB
- Available Space: 3.6 GB
- Disk Utilization: 48%

The root volume utilization is within acceptable limits and does not require immediate action.

7. Benefits of Using AWS Systems Manager

- No SSH or RDP access required.
- Improved security posture.
- Centralized management of EC2 instances.
- IAM-based access control.
- Ability to manage multiple instances simultaneously.
- Command execution history and auditing.
- Reduced operational overhead.

8. Limitations

- SSM Agent must be installed and running.
- IAM permissions must be configured correctly.
- Network connectivity to AWS Systems Manager endpoints is required.
- Primarily suitable for operational management rather than continuous monitoring.
- Additional monitoring solutions may be needed for long-term trend analysis.

9. Customer Communication Example

Scenario: High Disk Utilization

Subject: High Disk Utilization Detected on EC2 Instance

Dear Customer,

During our storage assessment, we observed that the EC2 instance is utilizing approximately 92% of its available storage capacity.

High disk utilization can lead to application performance issues, failed deployments, and service interruptions.

Recommended Actions:

- Remove unnecessary files and old backups.
- Configure log rotation for application logs.
- Increase the EBS volume size.
- Implement CloudWatch monitoring and alerts.
- Review storage usage patterns regularly.

Please let us know if you would like assistance implementing these recommendations.

10. Alternative Methods for Collecting Disk Utilization

Method 1: SSH Access

Administrators can connect directly to the EC2 instance using SSH and execute the `df -h` command to view disk utilization.

```
df -h
```

Method 2: Amazon CloudWatch Agent

The CloudWatch Agent can be installed on EC2 instances to collect disk utilization metrics and send them to CloudWatch.

Method 3: AWS Systems Manager Fleet Manager

Fleet Manager provides centralized visibility into managed instances and allows administrators to inspect instance details and perform management operations.